Advanced Architectures.

معماری پیشرفته

* معماری پیشرفته

معماری‌هایی که پیچیده‌اند و دارای فرمت‌های متغیر با بازه‌ی حجم کمتر

در مواقعی هستند که حافظه گران بوده از آن استفاده می‌کردند

Lenet 5

Dropout

BN

Relu

مقدار پارامترهای زیادی دارد.

پیشرو در زمان خودش

1 : conv

3 : conv

5 : conv

6 : FC

7 : FC

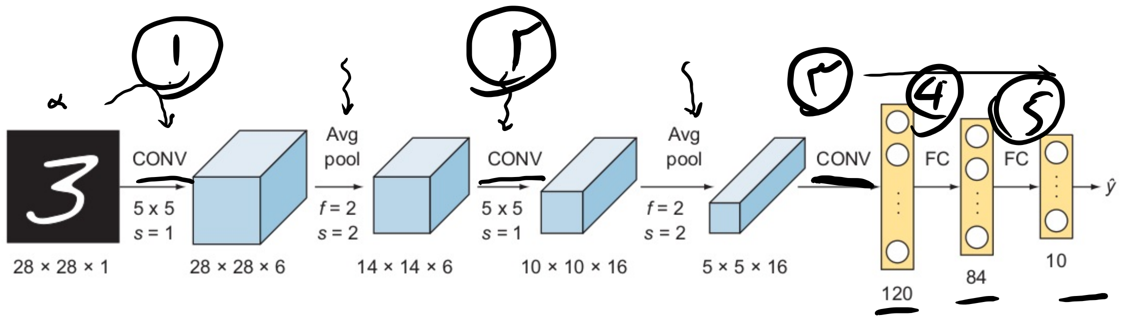


Figure 5.4 The LeNet architecture consists of convolutional kernels of size 5 × 5; pooling layers; an activation function (tanh); and three fully connected layers with 120, 84, and 10 neurons, respectively.

α ⟶ vanishing gradient ⟶ Relu

Alexnet

2012

تو رقابت‌ها اولین بار
Image class
کار با CNN بیشتر شد
روی تصاویر ...

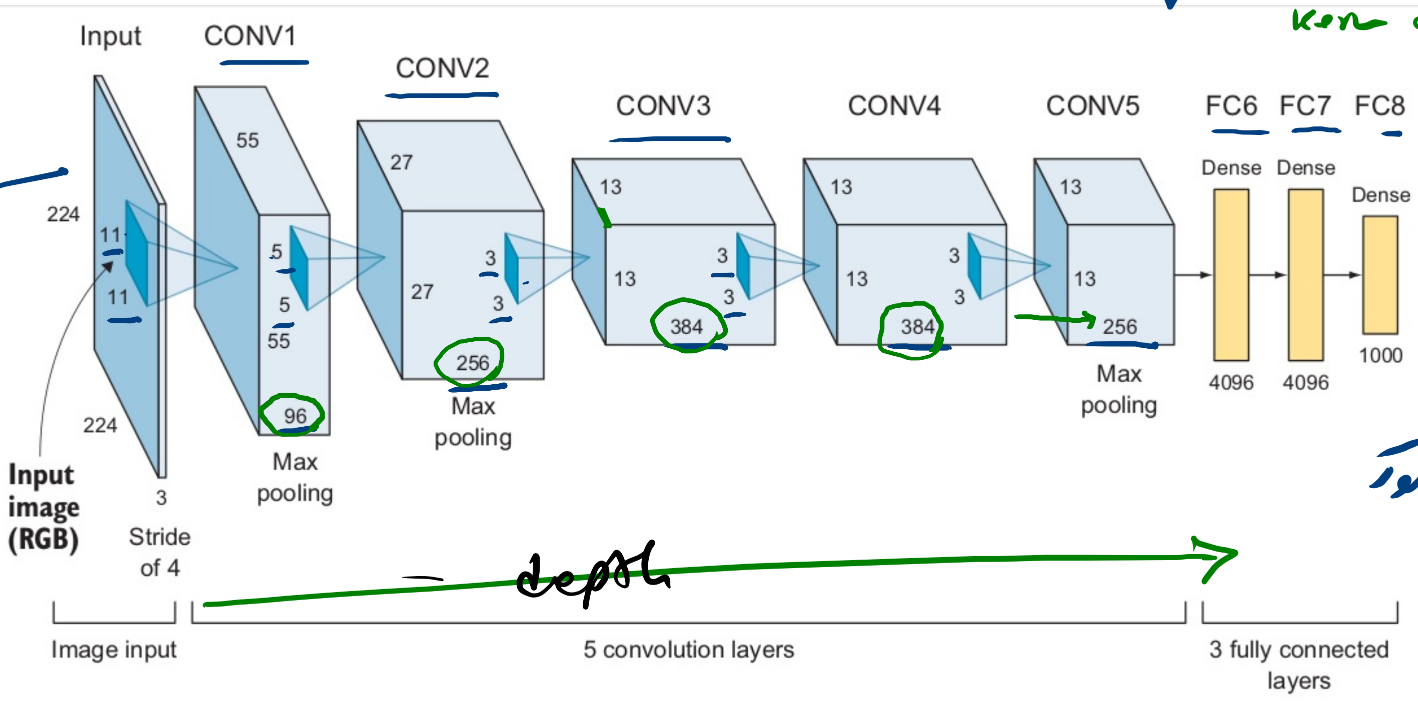
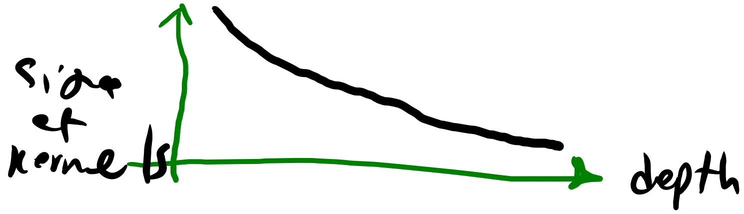


Figure 5.6 AlexNet architecture

* فیلترها رو اشتراک گذاشته توسط
* فیلتر 3x3 ثابت ، فیلتر 7x7 ، تصمیم گیری
مشابه!
خروجی عمیق



ILSVRC

Imagenet competition — .

1,2 m

1,000 class

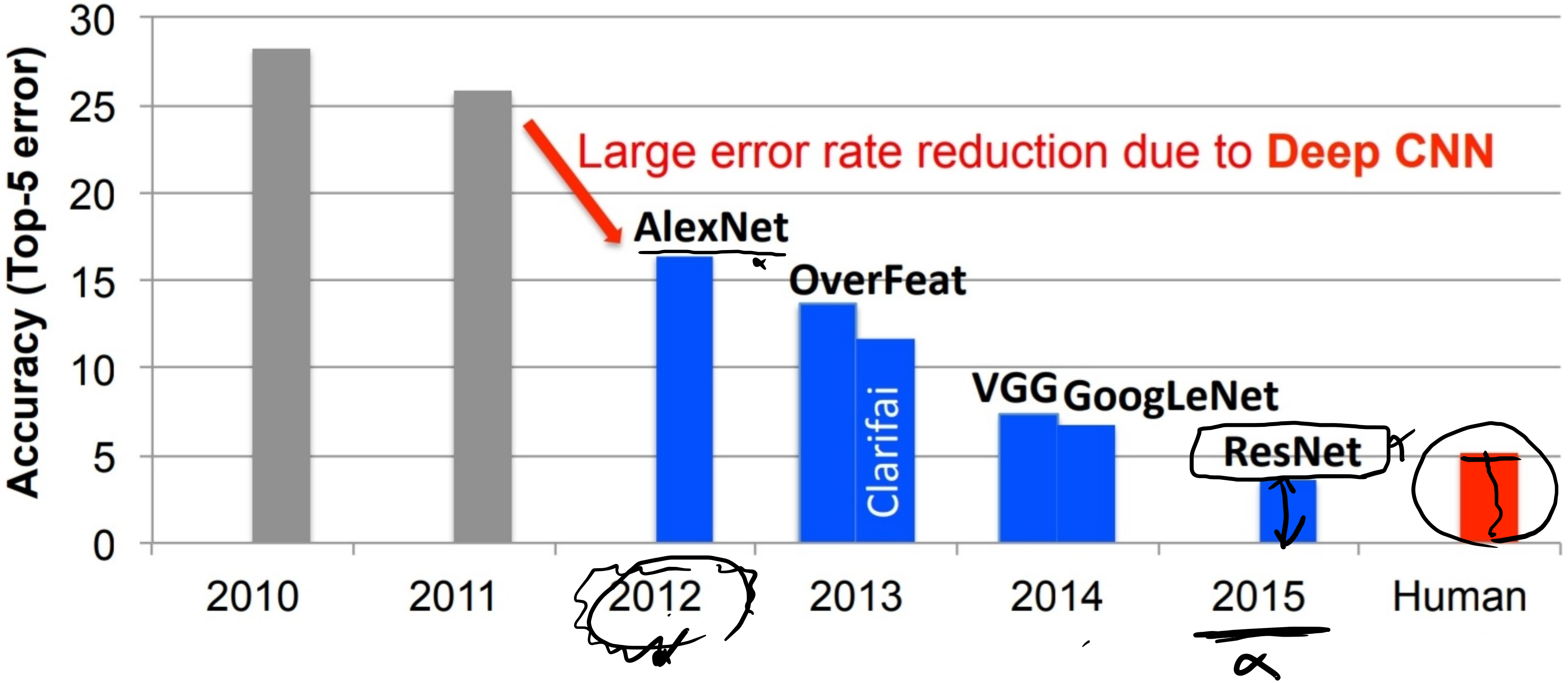


Fig. 7. Results from the ImageNet Challenge [14].

AI →

better Accuracy

more speed.

VGG16

2014

oxford

visual Geometry Group

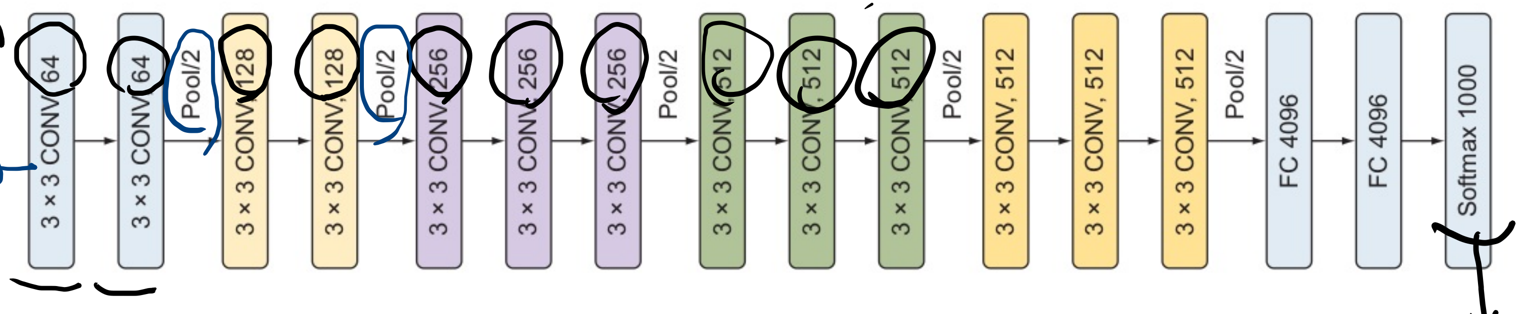


Figure 5.8 VGGNet-16 architecture

2014

GoogleNet

Inception module

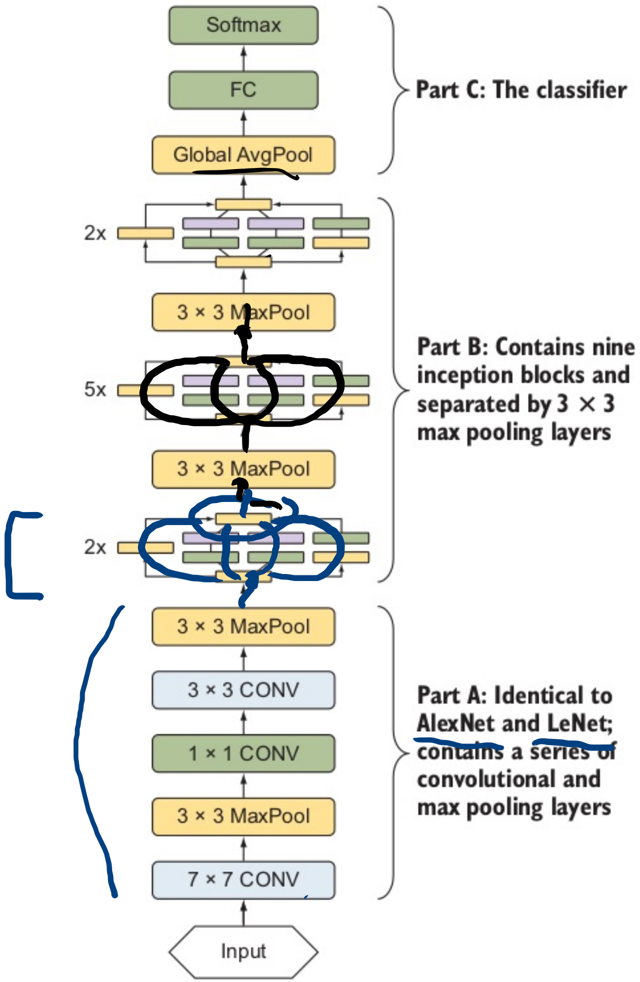


Figure 5.15 The full GoogLeNet model consists of three parts: the first part has the classical CNN architecture like AlexNet and LeNet, the second part is a stack of inceptions modules and pooling layers, and the third part is the traditional fully connected classifiers.

GoogleNet دقت بالاتری از VGG16 دارد! اما تا ۱۲ بار پارامترهای کمتری از

trainable parameters

VGG16 دارد.

why?

Inception Module.

Inception module

۱. حفظ ویژگی‌ها

۲. استخراج ویژگی‌های کلیدی

و بهتر!

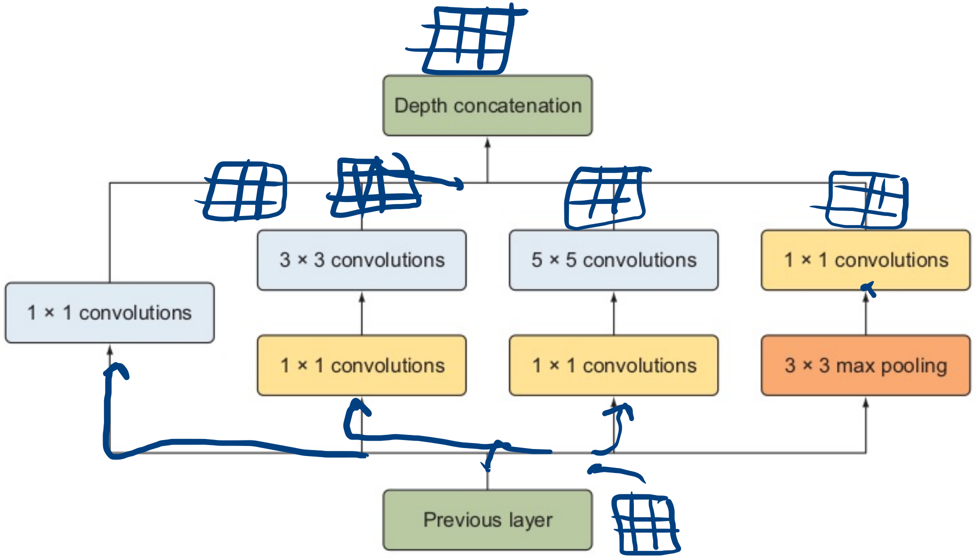
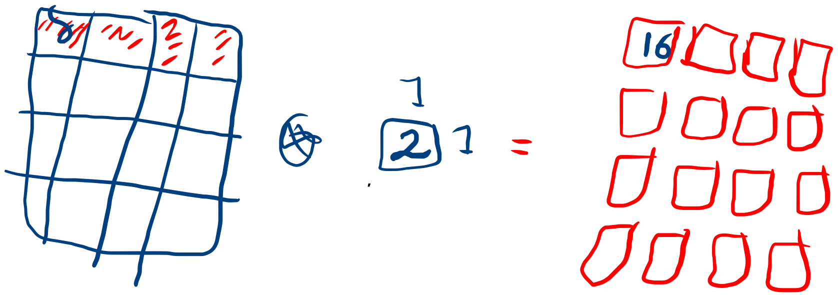
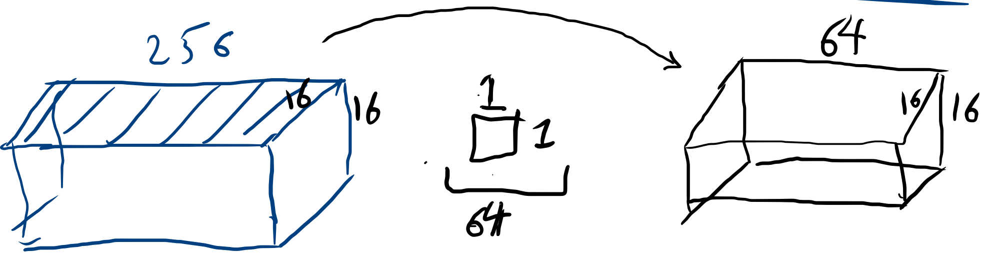


Figure 5.13 Building an inception module with dimensionality reduction





Conv(1x1, 64)

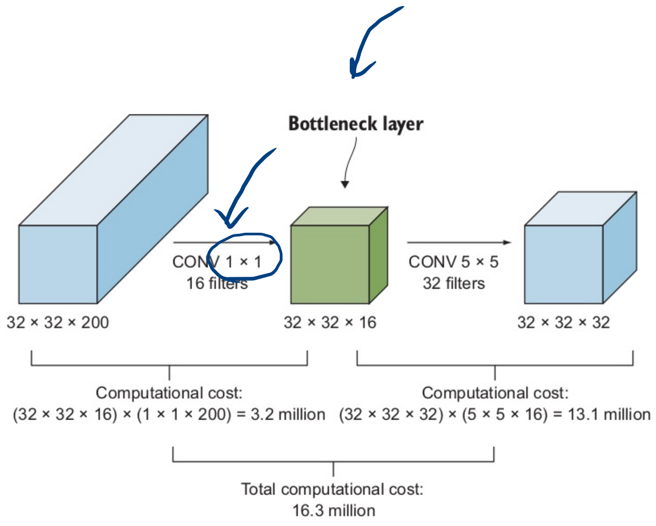
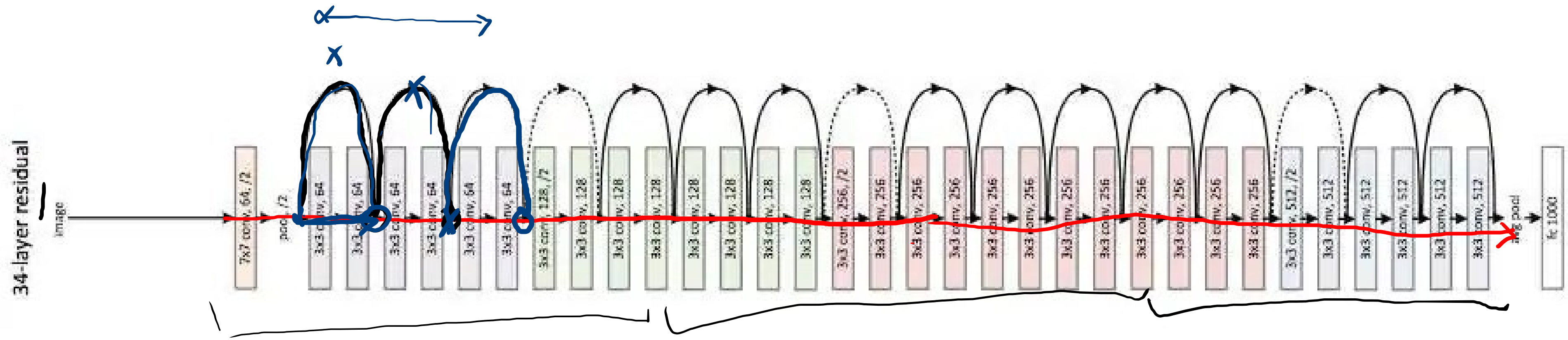


Figure 5.12 **Dimensionality reduction is used to reduce the computational cost by reducing the depth of the layer.**



Residual ← | میتونیم درطول شبکه حذف شوند!

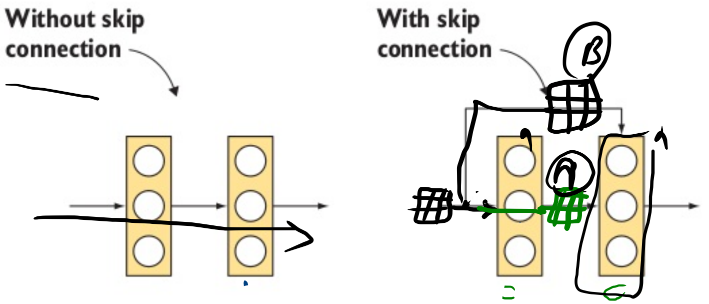


Figure 5.19 Traditional network without skip connections (left); network with a skip connection (right).

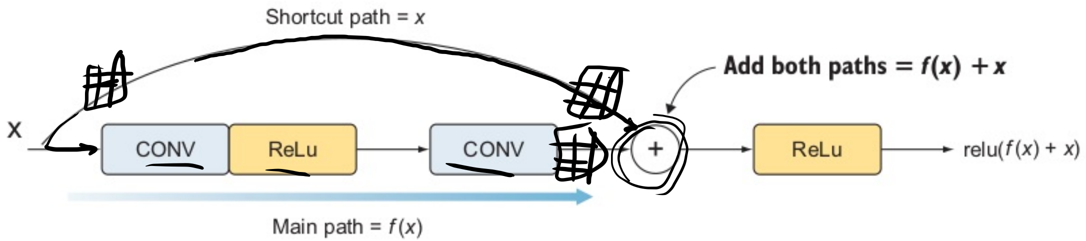


Figure 5.20 Adding the paths and applying the ReLU activation function to solve the vanishing gradient problem that usually comes with very deep networks

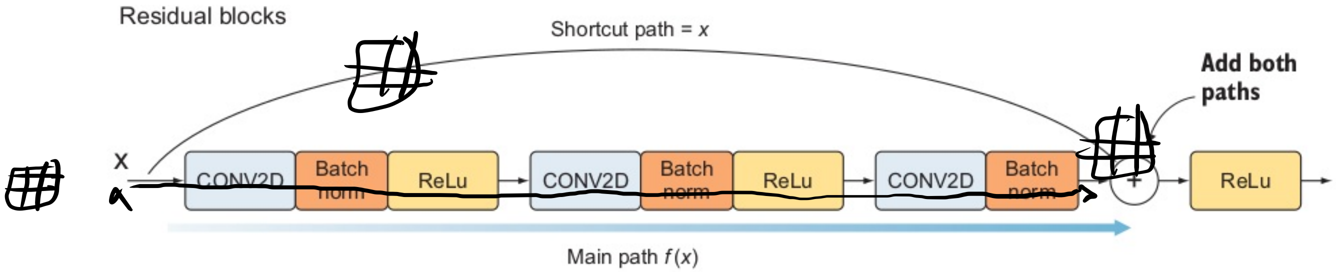


Figure 5.22 The output of the main path is added to the input value through the shortcut before they are fed to the ReLU function.

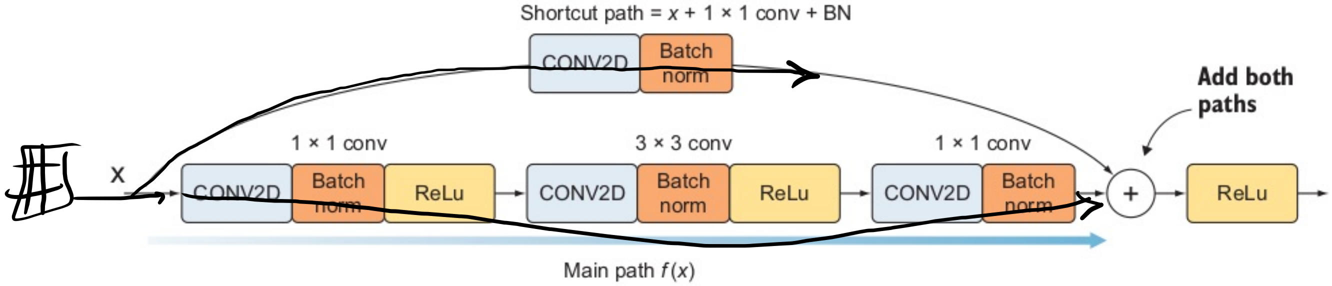


Figure 5.23 To reduce the input dimensionality, we add a bottleneck layer (1 × 1 convolutional layer + batch normalization) to the shortcut path. This is called the *reduce shortcut*.

googlenet

Resnet

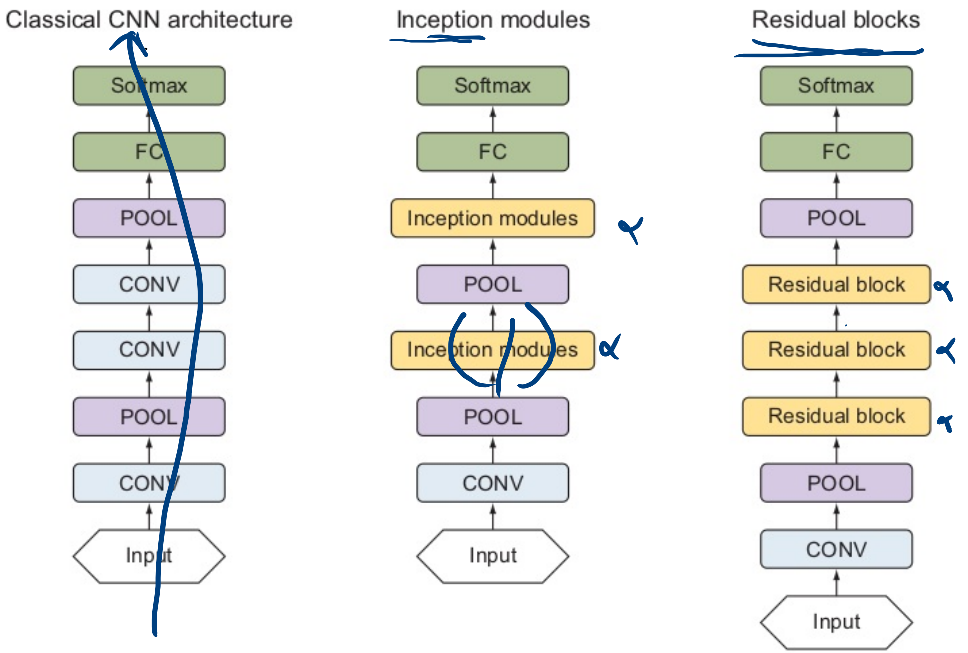


Figure 5.21 Classical CNN architecture (left). The Inception network consists of a set of inception modules (middle). The residual network consists of a set of residual blocks (right).

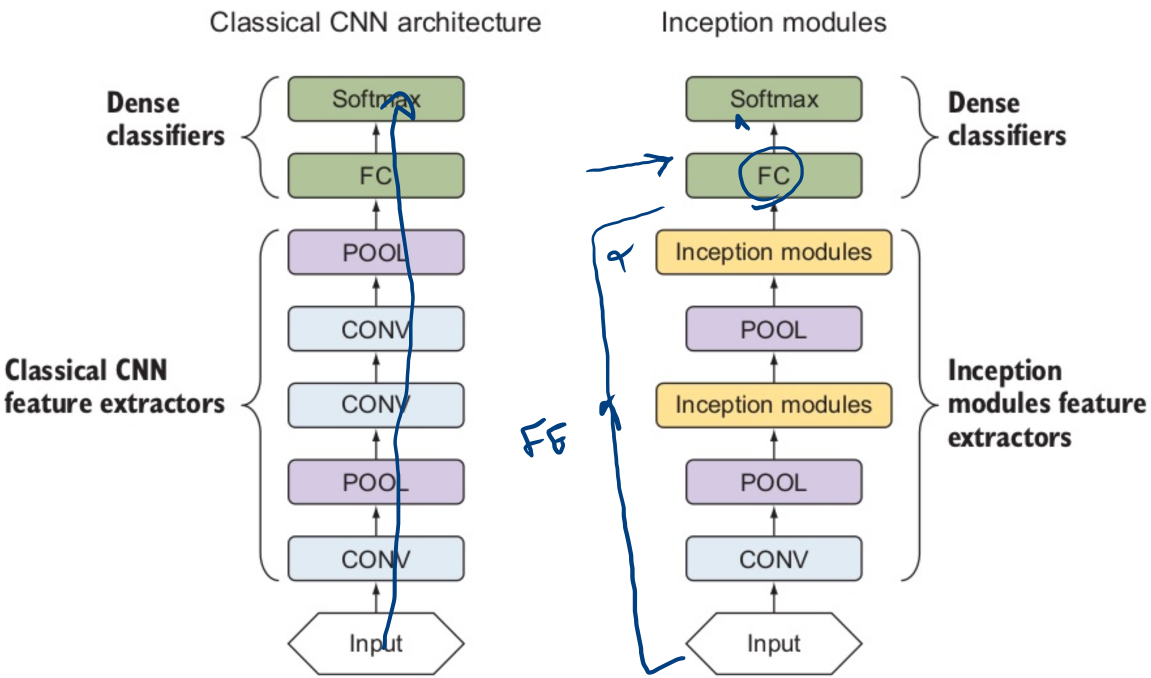


Figure 5.10 Classical convolutional networks vs. the Inception network

Lenet ✓

Alexnet ✓

Resnet ^ 16

googlenet ^

32

VGG16 → 64, 128 ...

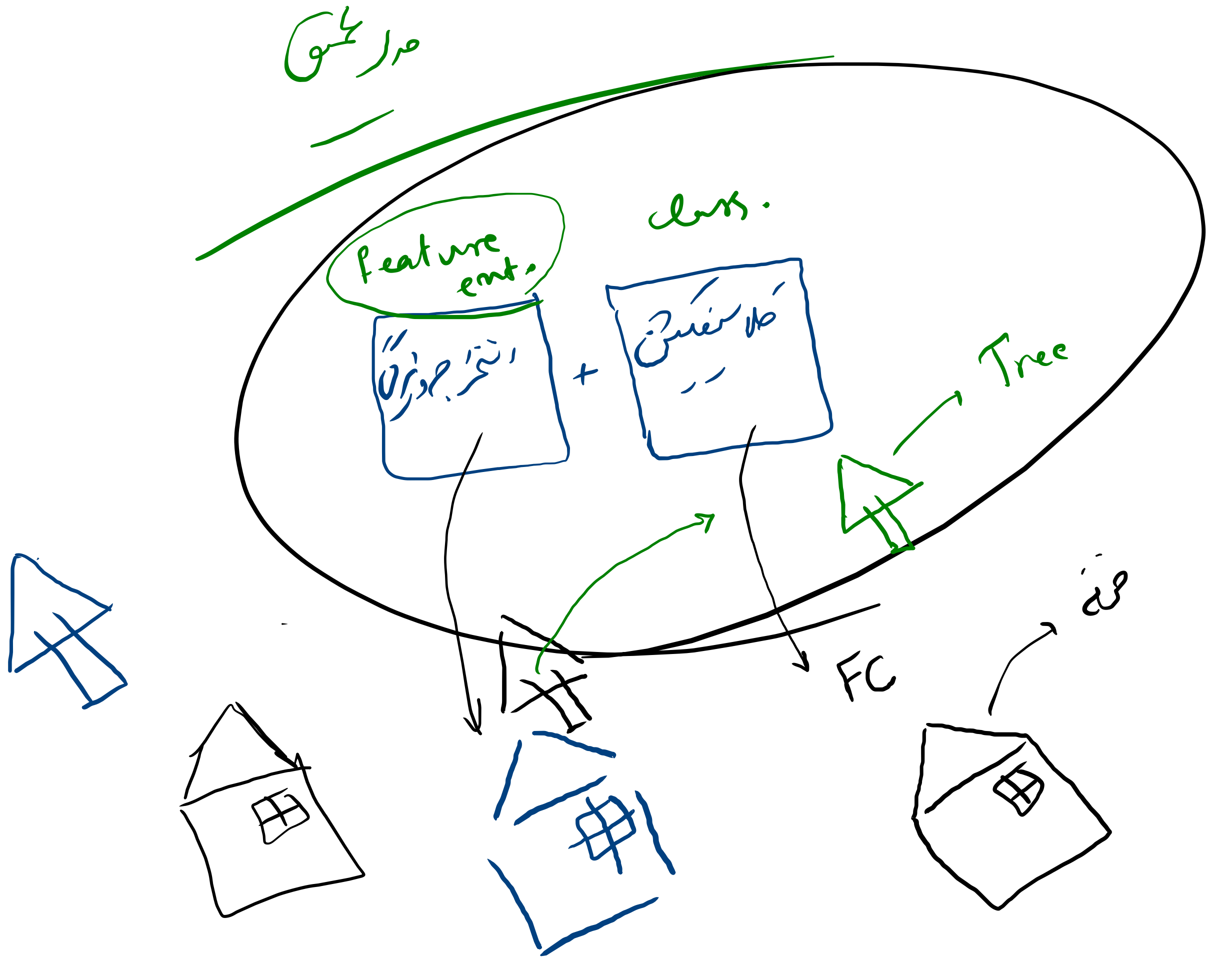
Advanced

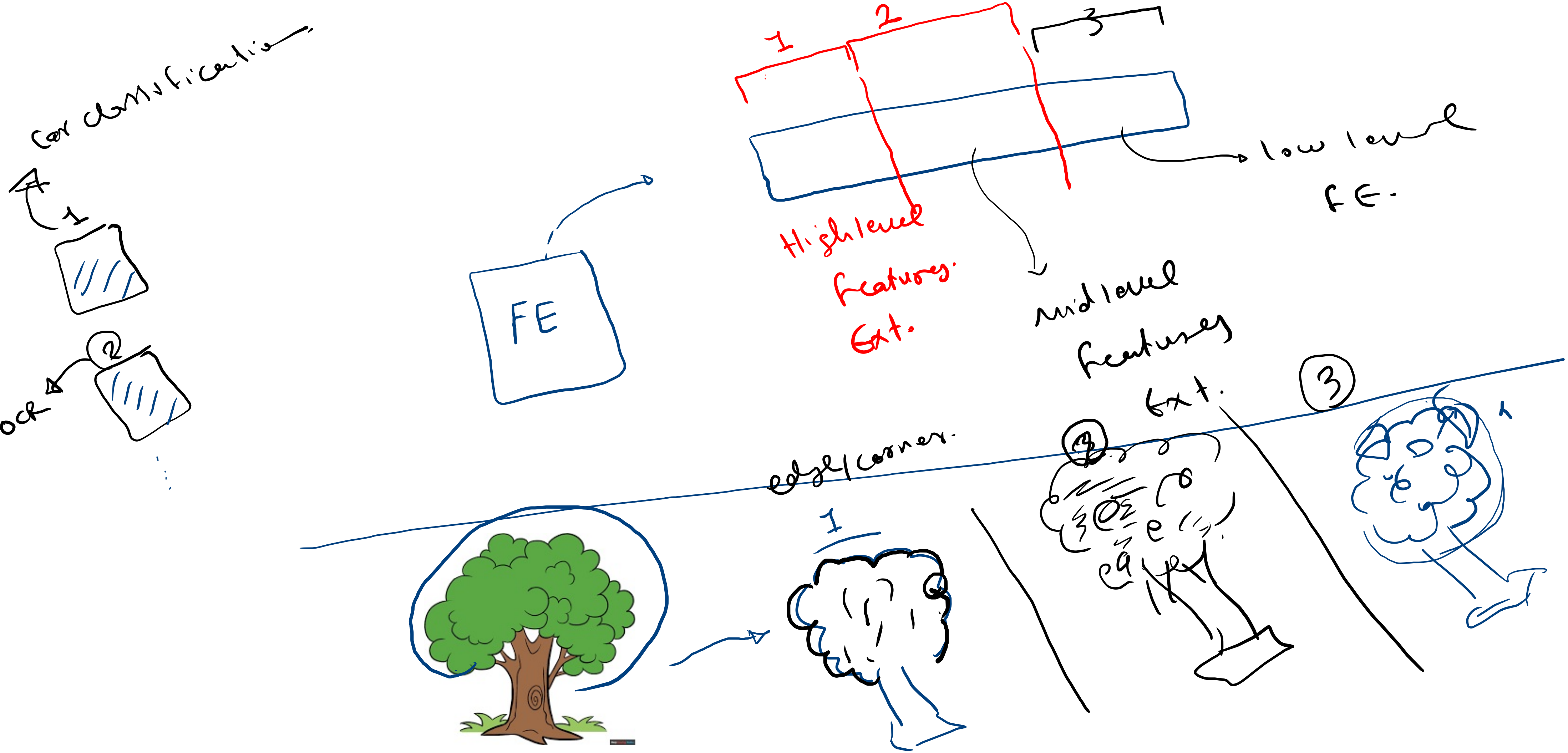
Architecture

* اگر من داده‌های زیادی برای آموزش شبکه عمیق نخواهند داشته باشم؟

اگر من ریسورس کافی برای آموزش شبکه‌های عمیق نخواهند داشته باشم؟

راه‌حل چیست؟


مدل عمیق
Feature ext.
class.
استخراج ویژگی
طبقه‌بندی
Tree
FC



Car classification
1
2
ock
FE
1
2
3
low level FE.
High level features Ext.
mid level features Ext.
edge/corner.
1
2
3

۵ مدل‌هایی که مدل‌های پایه را سرور اصلی (سرور اصلی) آموزش دیده‌اند
می‌توانند در استخراج ویژگی کاربرهای مختلف و به ما کمک کنند!

VGG 128 → GCP → 10,000,000 Image → Car classification

[illegible] BMW VW Hyundai

α α

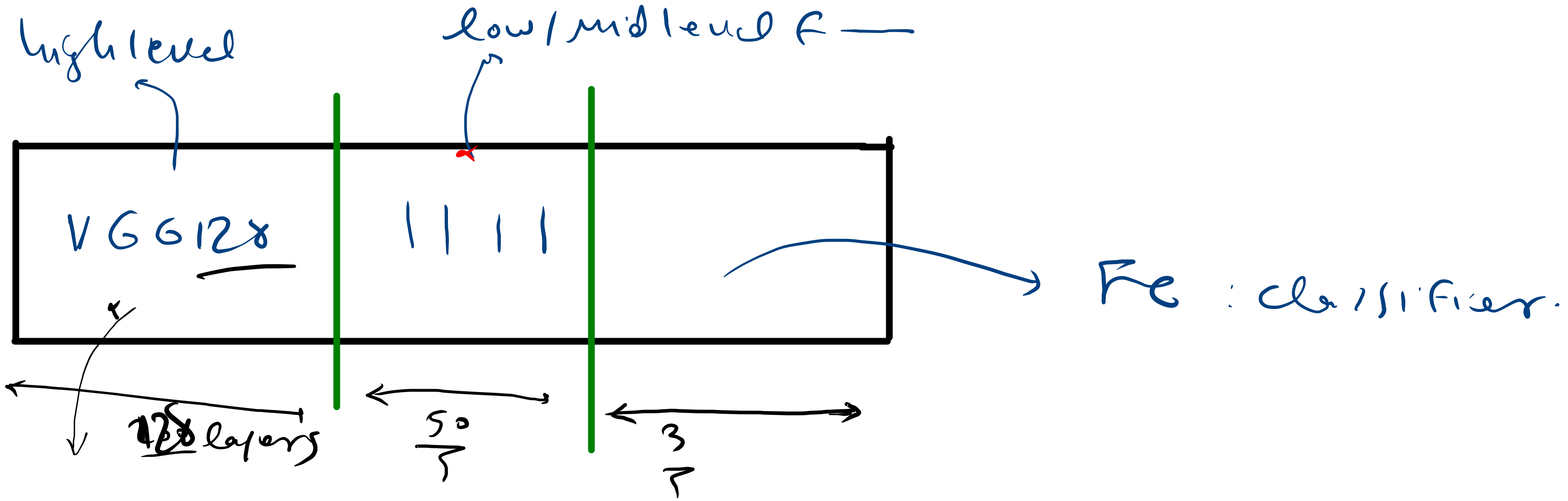
My Task : Gender

AI Model → Man / Woman

DS 10,000

high level

VGG 128 → my network → classification



Transfer learning!

128 layer : freeze.

50 + 3 : trainable layers.

Transfer
Learning /